

Table of Contents:

I. Appendix A

Methodological Deconstruction

*The Consultants name has been redacted for privacy and professional courtesy. The focus of this study is not either of us personally, but looks at larger issues that fall on each of us and the world.

II Appendix B

Table 1: Mean and Standard Deviation of Floral Features by Species (Summary Stats from initial EDA)

- **Table 1:** Mean and Standard Deviation of Floral Features by Species
- **Figure 1:** Pairwise scatterplots
- **Figure 2:** Scree Plot of Principal Components
- **Table 3:** Contingency Table of K-Means Cluster Assignments vs. Actual Species Labels

III Appendix C: Consultant Output & Diagnostics (Raw logs from script)

Logistic Regression Warning Logs

Model training and testing warning logs

Hyperlinks to client and consultants respective rmd and pdf/html outputs, and presentations.

Appendix A:

Methodological Deconstruction

Analysis Step	Consultant Approach (What Was Done)	Why It Was Problematic	What Should Be Done
Difference Testing	3 Linear Regressions (Sepal Length, Petal Length, Petal Width)	Conflated classification (regression) with inference (ANOVA); omitted the overlapping variable (Sepal Width); no multiple comparison correction.	MANOVA followed by protected ANOVAs & Tukey HSD to identify specific group differences.
Dimensionality	Regression on PC1 vs. Species	Circular reasoning; tests the summary statistic against the groups used to create it.	Use PCA for visualization/EDA only, not inferential testing of known labels.
Classification	Multinomial Logistic Regression	Ignored "complete separation" warnings; presented a broken model as valid; used a non-random train/test split.	Linear Discriminant Analysis (LDA) or Penalized Logistic Regression to handle separation.

Appendix B: Exploratory Data Analysis

Table 1: Mean and Standard Deviation of Floral Features by Species

Species	Sepal Length_Mean	Sepal. Length SD	Sepal. Width Mean	Sepal. Width SD	Petal. Length Mean	Petal. Length SD	Petal. Width Mean	Petal Width SD
Setosa	5.01	0.35	3.43	0.38	1.46	0.17	0.25	0.11
Versicolor	5.94	0.52	2.77	0.31	4.26	0.47	1.33	0.20
Virginica	6.59	0.64	2.97	0.32	5.55	0.55	2.03	0.27

Figure 1: Pairwise scatterplots visually highlighting the relationship between features (Petal length and width and sepal length and width).

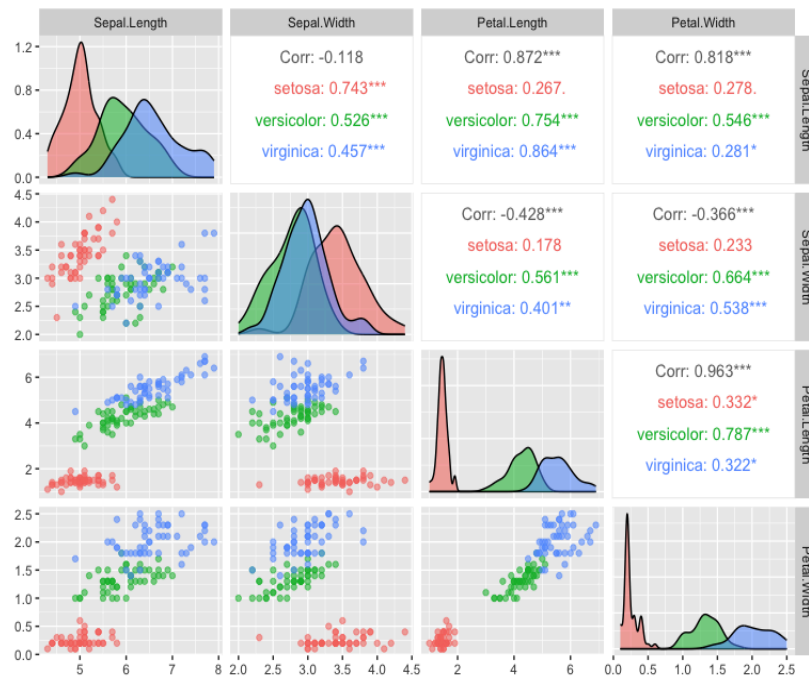


Figure 2. Scree Plot of Principal Components

Line plot of eigenvalues showing the variance explained by each component. The elbow after PC2 suggests a two-dimensional solution is appropriate.

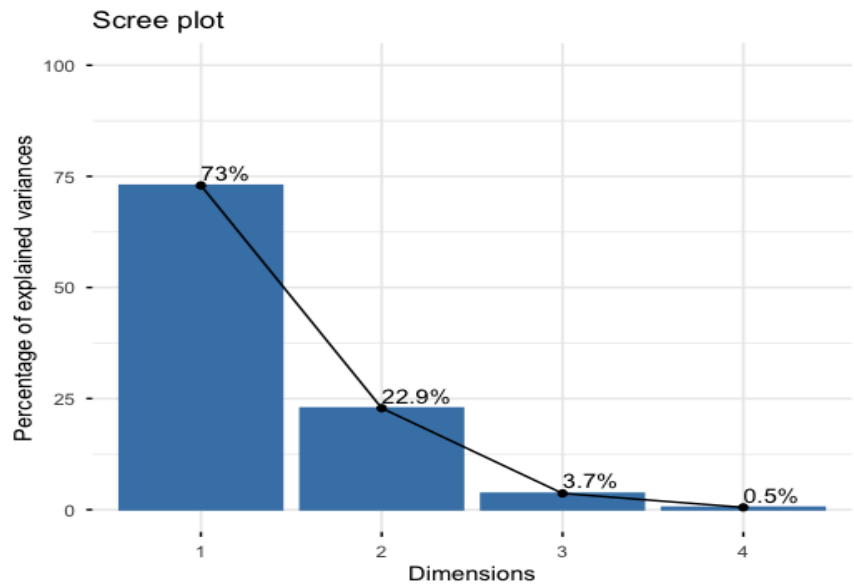


Table 3: Contingency Table of K-Means Cluster Assignments vs. Actual Species Labels

This table compares the clusters generated by K-means (k=3) with the true species labels in the Iris dataset. Each cell represents the count of observations falling into the intersection of a given cluster and species. Diagonal dominance suggests strong agreement between unsupervised clustering and ground truth class labels.

Table 3: Contingency Table – K-means Clusters vs. Species Labels

	Setosa	versicolor	virginica
0	0	48	14
0	0	2	36
50	50	0	0

Appendix C: Consultant Output & Diagnostics

- **Logistic Regression Warning Logs:** *Raw output generated from the consultant's code demonstrating model non-convergence.*

Multinomial Logistic Regression

```
``{r}
logistic_model <- vglm(
  Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width,
  family = multinomial,
  data = iris
)
Warning :2 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :13 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :22 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :34 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :39 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :41 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :47 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :50 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :54 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :59 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :63 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :78 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :91 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :96 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :fitted probabilities numerically 0 or 1 occurred
```

```

Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :97 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :some quantities such as z, residuals, SEs may be inaccurate due to convergence at a
half-step

```

```

      predict_class
      setosa versicolor virginica
setosa      50      0      0
versicolor   0     49      1
virginica    0      1     49

```

```

predict_probs <- predict(logistic_model, type = "response")
predict_class <- colnames(predict_probs)[apply(predict_probs, 1, which.max)]

```

```

table(iris$Species, predict_class)
```

```

## Model training and testing warning logs

# Training and Test Split

```

```{r}
test_index <- seq(1, 150, by = 3)

train <- iris[-test_index, ]
test <- iris[test_index, ]

logistic_train <- vglm(

```

```

Species ~ Sepal.Length + Sepal.Width + Petal.Length + Petal.Width,
family = multinomial,
data = train
)

predict_probs <- predict(logistic_train, newdata = test, type = "response")
predict_class <- colnames(predict_probs)[apply(predict_probs, 1, which.max)]

table(test$Species, predict_class)

```

...

```

Warning :1 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :5 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :11 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :18 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :24 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :28 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :33 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :34 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :38 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :40 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :42 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :52 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :61 diagonal elements of the working weights variable 'wz' have been replaced by
1.819e-12
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred
Warning :fitted values close to 0 or 1
Warning :fitted probabilities numerically 0 or 1 occurred

```

Warning :fitted values close to 0 or 1
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 Warning :fitted values close to 0 or 1
 Warning :fitted probabilities numerically 0 or 1 occurred
 Warning :fitted values close to 0 or 1
 Warning :some quantities such as z, residuals, SEs may be inaccurate due to convergence at a half-step

	predict_class		
	setosa	versicolor	virginica
setosa	17	0	0
versicolor	0	17	0
virginica	0	1	15

HYPERLINKS to rmd & PDFoutputs

- Consultant's full [RMD](#) and [PDF Yaml](#) Output
- Clients [initial rmd](#) & [supplemental reanalysis rmd](#)
- [Consultant_RedactedPresentation](#)
- [Txt](#) of Consultants silenced regression and model warnings